CAPSTONE / SYSTEM DOCUMENTATION

Updated System Flowchart & Architecture Diagram

NutriGoal — Smart Meal Planner

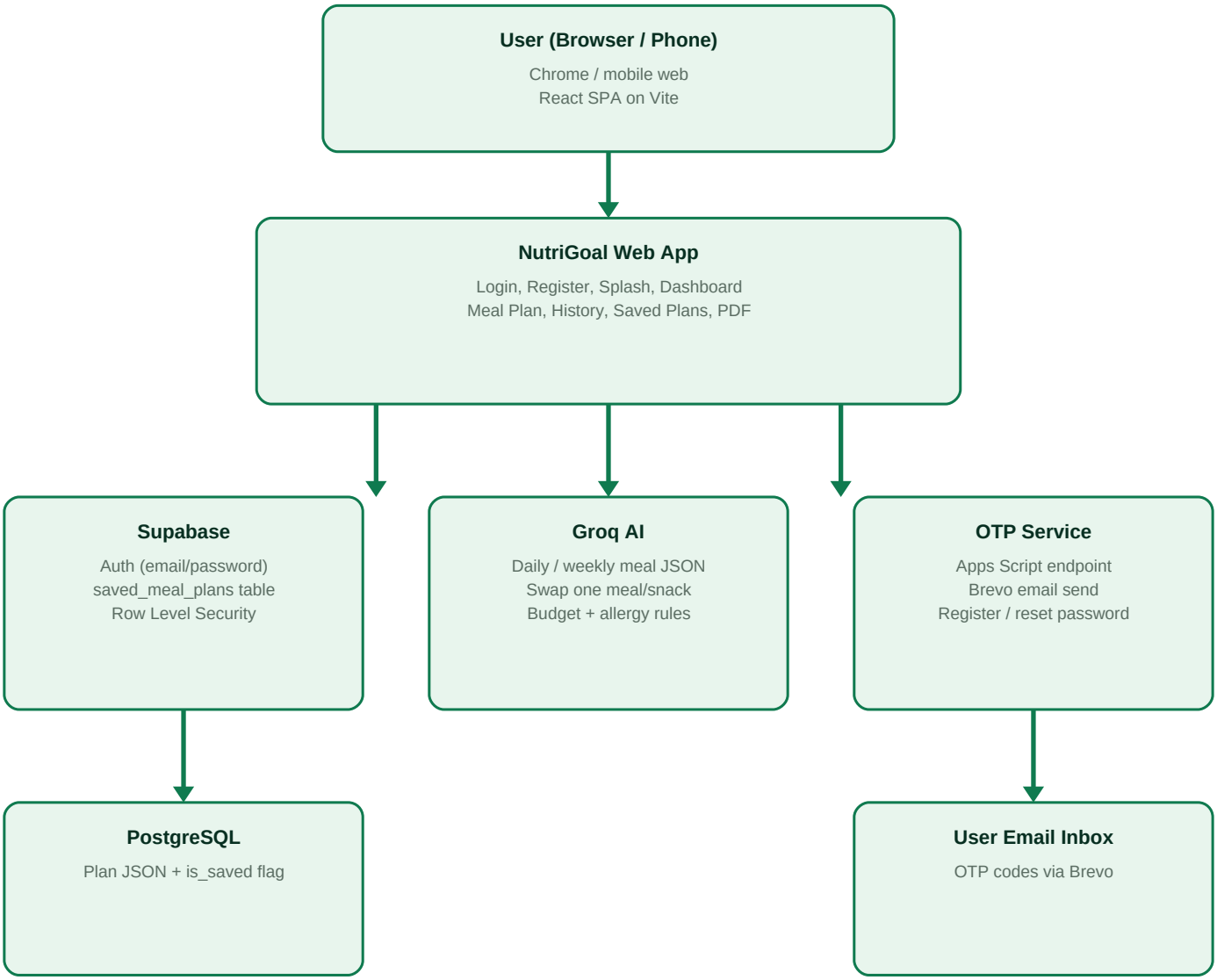
A web-based healthy meal recommender that uses user profile, budget, and AI to generate 1-day or 7-day Filipino meal plans.

Frontend	React 19 + Vite + React Router
Database / Auth	Supabase (PostgreSQL + Auth + RLS)
AI Generator	Groq API (openai/gpt-oss-20b)
Email OTP	Google Apps Script + Brevo
Export	jsPDF meal-plan download

Generated August 20, 2026

Based on the current NutriGoal source code.

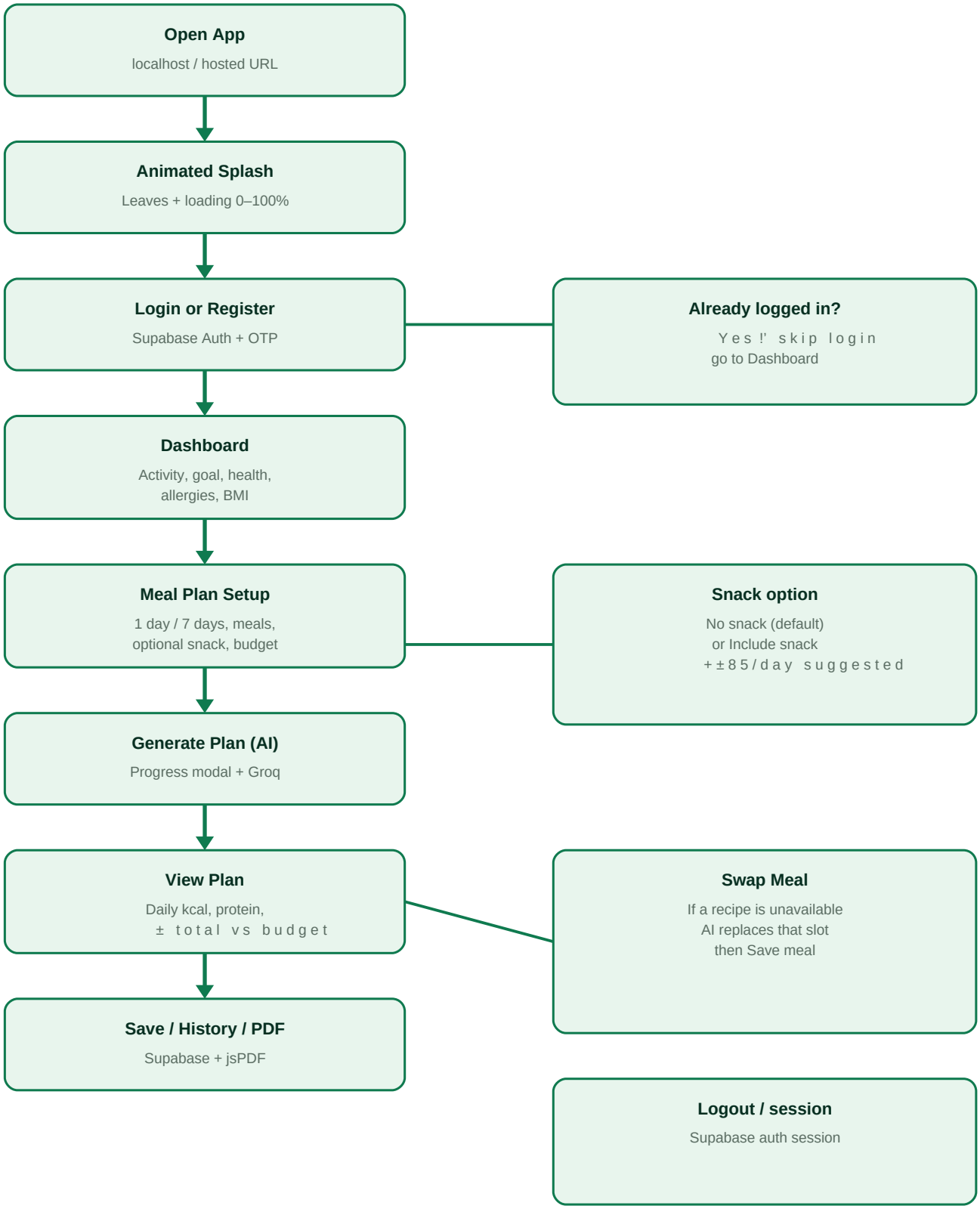
This diagram shows the main parts of NutriGoal and how they talk to each other. The browser never stores passwords in a custom table. Login is handled by Supabase Auth. Meal plans are saved as JSON in the saved_meal_plans table.



Request flow (simplified)

1. Guest opens /login → splash animation → login form.
2. After login, ProtectedRoute allows Dashboard, Meal Plan, History, Saved Plans.
3. Generate Meal Plan sends profile + budget + meals to Groq; fallback uses local recipes.
4. Generated plans are inserted into saved_meal_plans (history). Save marks is_saved = true.
5. Swap Meal calls Groq for one slot only, then updates that meal in the same plan row.
6. Register / forgot password uses the OTP Apps Script; Groq is not used for auth.

This is the main user journey. Each box is a screen or decision in the current app.



When the user taps Generate, NutriGoal builds a plan from profile filters, calorie target, and budget. Weekly plans call Groq once per day so the JSON does not get cut off.

A

Validate form

Meals (1–3), snack optional, budget "e minimum, BMI inputs, restrictions.

B

Compute targets

Daily calories from BMI/activity/goal. Budget split per meal; snack ~14% of daily budget.

C

Call Groq

Daily: 1 request. Weekly: 7 sequential day requests with progress (Planning Day X of 7).

D

Normalize JSON

Force Breakfast/Lunch/Dinner/Snack order. One recipe per meal. Realistic PHP prices.

E

Allergy + budget rules

Never use listed allergens. estimatedTotal must stay within each meal cap.

F

Fallback

If Groq fails/rate-limits, built-in Filipino recipes (localRecipes.js) are shown.

G

Save to History

Insert saved_meal_plans with is_saved = false. User can later Save to Saved Plans.

H

Swap (optional)

regenerateSingleMealWithGroq replaces one meal; updateMealPlan writes the same row.

Frontend screens

/login	Splash + Welcome back form
/register	Create account + OTP verify
/dashboard	Profile filters ⁉ recommended categories
/meal-plan	Generate 1-day or 7-day plan, snack, swap
/history	All generated plans
/saved-plans	Pinned plans + PDF download
/about	App information

Important source files

src/App.jsx	Routes, Guest vs Protected
src/login/LoginPage.jsx	Splash overlay then Login
src/mealfolder/Dashboard.jsx	Profile + category recommend
src/mealfolder/MealPlan.jsx	Generate, budget, snack
src/lib/groqMealPlan.js	AI generate + swap one meal
src/lib/mealPlanDb.js	Supabase insert/update/fetch
src/lib/budgetTier.js	Suggested budget + snack share
src/lib/otpClient.js	Apps Script OTP calls

Saved plan JSON (high level)

```
{ id, categoryTitle, planType: daily|weekly, meals[], includeSnack,
  budget, days[{ day, recommendations[{ meal, suggestedBudget, recipes[] }] }],
  filters: { activity, goal, allergies, BMI, protein, carb, ... }, source }
```

This document reflects the current NutriGoal implementation, including splash, snack, daily price totals, and swap meal.